CSR-MMDiff: Clear-Sky-Residual Diffusion for Calibrated Probabilistic PV Forecasting

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***Abstract*—Joint diffusion models of future sky images and photovoltaic (PV) power give strong probabilistic forecasts, but our compact 15-step re-implementation of PV-MM-Diffusion was under-dispersed, biased low and weaker than persistence as a point forecaster, and its generated frames blurred within minutes. We propose CSR-MMDiff, which diffuses only the *deviation* from a physical prior: PV power relative to clear-sky smart persistence and sky images relative to the last frame. It adds clear-sky conditioning, cloud-aware cross-modal attention, physically projected DDIM steps and a validation-fitted spread calibration. On the same 1,340 all-sky SKIPP'D test windows, a 30-member, 20-step ensemble of the $5.1\,\mathrm{M}$-parameter model, trained in $7.9\,\mathrm{h}$ on two RTX 3080 GPUs, reaches a CRPS of $0.943\,\mathrm{kW}$, which is 28.7% below smart persistence and 22.9% below our previous model. Its 90% intervals cover 83.2% of observations (previously 77.9%) while being narrower, its median beats persistence (MAE 1.243 vs. $1.322\,\mathrm{kW}$), generated frames keep an SSIM of 0.724 at +16 min (previously 0.423), and sampling takes $54\,\mathrm{ms}$ per member. On a cloudy subset that mirrors the PV-MM-Diffusion protocol, the skill is 25.4%. Coverage remains below nominal, and on the most variable days the median lags ramps.**

***Index Terms*—solar forecasting, photovoltaic power, sky images, diffusion models, DDIM, clear-sky model, probabilistic forecasting, calibration**

I. Introduction

Ultra-short-term PV forecasting is increasingly multimodal. PV-MM-Diffusion [1] (B3) denoises future sky frames and PV power jointly in one reverse process on the SKIPP'D benchmark [2]; PARA-PV [3] adds physics-aware retrieval with a frozen foundation model, Jiang et al. [4] fuse sky images with time series, and SkyEvo [5] forecasts from generated sky-image evolution. A passing cloud can halve the output of a PV plant within a minute, such ramps are invisible to numerical weather prediction [6], and sky cameras have become the standard nowcasting input [7], [8]. Grid operators need the *distribution* of future output [9], while squared-error regressors collapse to a smoothed persistence [10]; generative models such as SkyGPT [11] instead sample plausible futures.

We previously trained two models of this class ourselves and use them as baselines. B1, a compact 15-step DDIM re-implementation of PV-MM-Diffusion, beat persistence in CRPS, but its 90% intervals covered 77.9% of observations, its median was biased low and worse than persistence, and its frames lost structure within minutes. B2, a larger joint model, showed that 10-step DDIM cut latency 100-fold relative to 1000-step DDPM but reduced 90% coverage from 87.5% to 64.8%. Both models spend capacity re-generating what is already known: the diurnal PV cycle and the static sky. Our contributions are:

- *CSR-MMDiff*, a **clear-sky-residual** diffusion target that makes smart persistence and the last frame the zero point of the model, with **physics-informed conditioning** on clear-sky power known ahead and modality flags;
- a **cloud-aware RS-PV-MMA** with learned spatial pooling, and a **projected DDIM** sampler with **lead-wise spread calibration**;
- a like-for-like comparison with B1 on its 1,340 test windows and a protocol-matched one with B2 and B3, covering probabilistic and point accuracy, calibration and image quality.

II. Related Work

Sky-image regressors such as SUNSET [7] predict PV from stacked images and PV history. SkyGPT [11] and SkyEvo [5] generate future skies and map them to PV in a second stage, whereas PV-MM-Diffusion [1] adapts MM-Diffusion [12] to generate both modalities at once and outperforms SkyGPT→U-Net. Diffusion [13] also serves video [14], time series [15], [16] and precipitation nowcasting [17], [18]; DDIM [19] and DPM-Solver++ [20] shorten sampling. Proper scores such as CRPS must be complemented by coverage diagnostics [21], [22]. CSR-MMDiff instead places a physical forecast *inside* the diffusion target.

III. Method

A. Clear-Sky-Residual Formulation

Let $\mathbf{I}_t \in \mathbb{R}^{64\times64\times3}$ be the sky image and $P_t \in [0, P_{\max}]$ the PV power ($P_{\max} = 30\,\mathrm{kW}$), sampled every 2 min. Given the history $\mathcal{H} = \{\mathbf{I}_{t-2j}, P_{t-2j}\}_{j=0}^{7}$, the model samples the future $\{\mathbf{I}_{t+k}, P_{t+k}\}$, $k \in \{2,\dots,16\}$ min. The clear-sky power P^{clr} is the pvlib Ineichen clear-sky irradiance [23] scaled to $P_{\max}$; the time zone of the unlabelled SKIPP'D timestamps is chosen by maximum correlation with the measured PV (America/Los_Angeles, $\rho = 0.881$). Smart persistence (SPM) holds the clear-sky index constant,

$$\hat{P}^{\mathrm{SPM}}_{t+k} = k_t\, P^{\mathrm{clr}}_{t+k}, \qquad k_t = \mathrm{clip}\big(P_t/P^{\mathrm{clr}}_t,\, 0,\, 2.5\big), \tag{1}$$

and falls back to plain persistence when $P^{\mathrm{clr}}_t < 1\,\mathrm{kW}$. Instead of the raw future, CSR-MMDiff diffuses the residuals

$$x^{P}_{0,k} = \frac{P_{t+k} - \hat{P}^{\mathrm{SPM}}_{t+k}}{s_P}, \qquad \mathbf{x}^{I}_{0,k} = \frac{\mathbf{I}_{t+k} - \mathbf{I}_t}{s_I}, \tag{2}$$

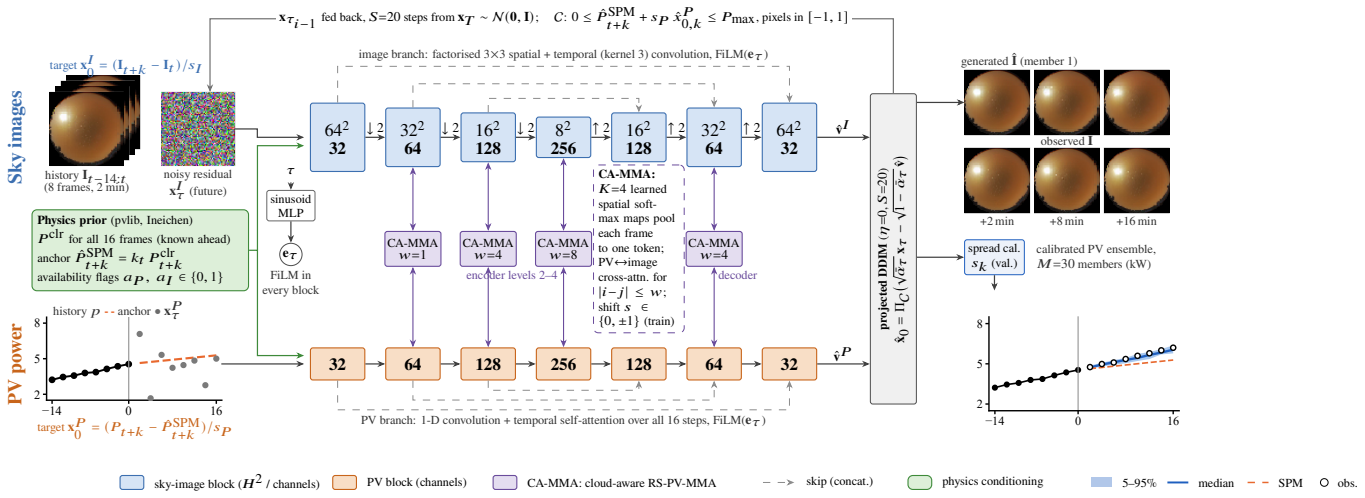


Fig. 1. CSR-MMDiff on a real test window (2019-07-14 08:12, also used to inspect B1). Both branches diffuse residuals: frames relative to the last frame, PV relative to the clear-sky smart-persistence anchor (orange); the physics prior and flags are extra input channels. Cloud-aware RS-PV-MMA (CA-MMA) couples the branches. Outputs after 20 projected DDIM steps: generated (member 1) vs. observed frames, and the calibrated 30-member PV forecast (band 5–95%, median, observed circles).

with $s_P = 3.96\,\mathrm{kW}$ and $s_I = 0.165$ chosen on the training windows to give the targets a standard deviation of 0.5. The zero residual thus equals persistence, and the network only has to model cloud-induced deviations. When the PV history is withheld, the anchor becomes $\bar{k}\,P^{\mathrm{clr}}_{t+k}$ with the climatological index $\bar{k} = 0.842$.

B. Physics-Informed Conditioning

Each of the 16 frames carries seven PV channels (normalized PV history, noisy residual, future flag, $P^{\mathrm{clr}}/P_{\max}$ known for past *and* future frames, level given by history or anchor, and availability flags a_P, a_I) and eight image channels (history frame or anchor frame $\mathbf{I}_t$, noisy residual, future flag, a_I). The flags remove an ambiguity of B1, whose zero placeholder for a missing modality coincided with 15 kW in its $[-1, 1]$ scaling. As in [1], training windows use both modalities, images only or PV only (probabilities 0.7/0.15/0.15).

C. Coupled U-Net with Cloud-Aware RS-PV-MMA

The denoiser (Fig. 1) has two four-level branches, widths $(32, 64, 128, 256)$ and FiLM [24] timestep modulation; image blocks factorize spatial and temporal convolutions, PV blocks add temporal self-attention [25]. B1 and B2 pool each image frame to one token by global averaging, which discards *where* clouds are relative to the sun. CA-MMA instead learns $K = 4$ spatial attention maps over the positions of the feature map $\mathbf{h}_t$:

$$\begin{aligned} \mathbf{A}_t &= \mathrm{softmax}_{hw}\big(\mathbf{W}\mathbf{h}_{t,hw}\big), \\ \mathbf{z}_t &= \Big\|_{\kappa=1}^{K} \textstyle\sum_{hw} A_{t,hw\kappa}\, \mathbf{h}^{(\kappa)}_{t,hw}, \end{aligned} \tag{3}$$

where $\mathbf{h}^{(\kappa)}$ is the κ-th channel group and $\mathbf{W}$ is zero-initialized, so training starts from average pooling. Cross-attention between the PV and image tokens is restricted to $|i - j| \le w$ with $w = 1, 4, 8$ at encoder levels 2–4; a fourth fusion ($w = 4$) at decoder level 2 passes features of the generated frames to the PV head. Image tokens are rolled by $s \in \{-1, 0, 1\}$ during training, and output projections are zero-initialized. The model has 5.09 M parameters.

D. Training Objective

We use v-prediction [26] on a cosine schedule [27] with $T = 1000$. With $\mathbf{x}_\tau = \sqrt{\bar{\alpha}_\tau}\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_\tau}\boldsymbol{\epsilon}$ and $\mathbf{v} = \sqrt{\bar{\alpha}_\tau}\boldsymbol{\epsilon} - \sqrt{1-\bar{\alpha}_\tau}\mathbf{x}_0$, the loss is

$$\mathcal{L} = \mathbb{E}\Big[\omega_\tau\big(\|\mathbf{v}^P - \hat{\mathbf{v}}^P\|^2 + \lambda_I\, a_I\, \|\mathbf{v}^I - \hat{\mathbf{v}}^I\|^2\big)\Big], \tag{4}$$

with Min-SNR weight $\omega_\tau = \min(\mathrm{SNR}_\tau, 5)/(\mathrm{SNR}_\tau + 1)$ [28], $\lambda_I = 0.5$, and the image loss masked when images are withheld.

E. Projected DDIM Sampling and Spread Calibration

Sampling starts from $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ and runs $S = 20$ DDIM [19] steps. At every step the clean estimate is projected onto the feasible set $\mathcal{C}$:

$$\hat{\mathbf{x}}_0 = \Pi_{\mathcal{C}}\big(\sqrt{\bar{\alpha}_\tau}\mathbf{x}_\tau - \sqrt{1-\bar{\alpha}_\tau}\hat{\mathbf{v}}\big), \tag{5}$$

where $\mathcal{C}$ requires $0 \le \hat{P}^{\mathrm{SPM}}_{t+k} + s_P \hat{x}^P_{0,k} \le P_{\max}$ and all pixels to lie in $[-1, 1]$. The noise estimate is recomputed from the projected $\hat{\mathbf{x}}_0$ before the update. The ensemble $\{\hat{P}^{(m)}_{t+k}\}_{m=1}^{M}$ with median $\tilde{P}_{t+k}$ is then recalibrated per lead time,

$$\hat{P}'^{(m)}_{t+k} = \mathrm{clip}\big(\tilde{P}_{t+k} + s_k(\hat{P}^{(m)}_{t+k} - \tilde{P}_{t+k}),\, 0,\, P_{\max}\big), \tag{6}$$

where $s_k \in [0.5, 3]$ minimizes the CRPS on 1,536 validation windows. The median, and hence every point metric, is unchanged. We also evaluated stochastic sampling ($\eta > 0$) and a sky-image guidance $\hat{\mathbf{v}}^P = \hat{\mathbf{v}}^P_{\mathrm{SI+PV}} + g(\hat{\mathbf{v}}^P_{\mathrm{SI+PV}} - \hat{\mathbf{v}}^P_{\mathrm{PV}})$, which the placeholder training makes available at no cost.

IV. Experimental Setup

Data and protocol. SKIPP'D [2] pairs 64×64 sky images at 1-min resolution with the power of a 30 kW array at Stanford. As in B1, 16-frame windows at 2-min cadence never cross days or logging gaps; training and validation are split by calendar day (447/50 days, 99,473/11,096 windows), and the test set comprises the 1,340 windows used to evaluate B1on 20 clear, partly cloudy and overcast days. To mirror B3, which tests on five multi-cloud days with clear-sky samples removed, we also form a *cloudy subset* from the five most variable test days (2017-09-20, 2018-01-25, 2018-10-04, 2019-05-27, 2019-10-19). It keeps daytime windows at 2-min stride that contain a 2-min PV change of at least 1 kW, which gives 1,199 windows.

Baselines. SPM; a probabilistic persistence ensemble (PeEn) that adds training-residual quantiles, binned by lead time and clear-sky power, to SPM; our earlier joint DDIM B1 and its two-stage video→PV pipeline on the identical windows; our earlier DDIM ensembles B2; and PV-MM-Diffusion and SkyGPT→U-Net as reported in B3. The same trained model is also sampled with one modality withheld.

Implementation. TensorFlow 2.21 on two NVIDIA RTX 3080 GPUs (10 GB each; data-parallel, fp16, peak 5.8 GB per GPU), Adam [29] (learning rate 2×10^{-4}, cosine decay), batch size 32 and EMA weights (0.9995). Training stopped early after 75,000 steps (7.9 h); the best EMA checkpoint by validation CRPS (step 45,000) is used. η, g, S and s_k are chosen on validation days only; test ensembles have $M = 30$ members (seed 42).

Metrics. The empirical CRPS $= \frac{1}{M}\sum_m |\hat{y}^{(m)} - y| - \frac{1}{2M^2}\sum_{m,m'} |\hat{y}^{(m)} - \hat{y}^{(m')}|$, averaged over windows and leads, equals the MAE for deterministic forecasts; skill is $1 - \mathrm{CRPS}/\mathrm{CRPS}_{\mathrm{SPM}}$. The central 90% interval is scored by the Winkler score WS_{90}, coverage and width [9], the median by MAE, RMSE, mean bias error (MBE) and R^2. Generated frames (first member) are compared with the observed frames by SSIM and by the cosine similarity of VGG-16 features (VGG-CS) [30], [31].

V. Results

Probabilistic accuracy. Table I compares all methods on identical windows. CSR-MMDiff lowers the CRPS of the B1 joint model from 1.222 to 0.943 kW (−22.9%) and its WS_{90} from 10.50 to 9.15 kW. Skill over SPM rises from 7.5% to 28.7%, and the CRPS is 20% below PeEn. The gain holds at every lead (Fig. 2(a)), from 23% skill at +2 min, where B1 was worse than SPM, to about 30% beyond +10 min. Withholding either modality removes the gain (PV only: 1.579 kW, with intervals inflated by the full model's spread factors; sky images only: 1.373 kW), so both are needed.

Calibration. The raw ensemble is sharp but underdispersed: its 90% intervals cover only 57.4% of observations. The fitted spread factors are almost constant ($s_k = 2.20, 2.25, 2.20, 2.15, 2.00, 2.05, 2.00, 1.95$ from +2 to +16 min), which points to a systematic rather than lead-dependent deficit. Calibration raises coverage to 83.2%, lowers WS_{90} by 25.8% and the CRPS by 5.2%. Coverage stays between 81% and 85% at every lead (Fig. 2(b)), whereas B1 decayed from 81.6% to 75.1%. Our intervals are also *narrower* (2.7–4.7 kW vs. 3.7–6.1 kW; Fig. 2(c)), i.e., sharper and more reliable, though still 6.8 points below nominal.

TABLE I
Accuracy on the 1,340 SKIPP'D Test Windows (2–16 min) of B1. Skill vs. Smart Persistence; Nominal Coverage 90%.

Method	CRPS (kW)	Skill (%)	WS_{90} (kW)	Cov. (%)	MAE (kW)	RMSE (kW)	MBE (kW)
Smart persistence†	1.322	0.0	–	–	1.322	3.087	−0.04
Two-stage B1†	1.498	−13.4	–	–	1.498	**2.838**	0.17
PeEn	1.177	11.0	18.92	76.9	1.315	3.087	0.04
Joint DDIM v1 B1	1.222	7.5	10.50	77.9	1.684	2.960	−0.76
Ours, PV only	1.579	−19.4	17.20	99.6	1.568	2.864	0.60
Ours, SI only	1.373	−3.9	12.60	89.4	1.812	3.335	0.53
Ours, uncalibrated	0.995	24.8	12.33	57.4	**1.243**	2.878	0.21
Ours (CSR-MMDiff)	**0.943**	**28.7**	**9.15**	**83.2**	**1.243**	2.878	0.21

†Deterministic: CRPS equals MAE and interval scores are undefined. B1 rows from our earlier evaluation on the same windows; R^2: SPM 0.824, two-stage 0.851, B1 0.838, ours 0.847.

TABLE II
Comparison under Different Test Protocols. Each Uses Its Own Test Days and SPM; Only Relative Skill Is Comparable.

Method	Test protocol	CRPS (kW)	WS_{90} (kW)	Cov. (%)	Skill (%)
PeEn	5 variable days,	3.488	58.00	38.5	8.9
Ours (CSR-MMDiff)	1,199 cloudy win.	**2.856**	26.95	**76.1**	25.4
SkyGPT→U-Net [11]	B3: 5 cloudy	3.21	35.62	–	12.5
PV-MM-Diffusion B3	days, 2,582 win.	**2.63**	**21.46**	–	**28.3**
DDIM S=10, η=0.5 B2	20 days, +16 min	1.144	11.80	≈70	41.4*
Ours at +16 min	20 days, +16 min	1.198	–	83.1	29.7

Skill vs. each study's SPM (B3: 3.67 kW; ours, cloudy: 3.830 kW). *6,627 windows, solar-geometry SPM surrogate; 64.8% coverage on B2's matched subset (DDPM: 87.5%).

Point accuracy. Because the zero residual equals SPM, the median no longer falls behind persistence: its MAE of 1.243 kW is 6.0% below SPM and 26.2% below B1, and the low bias of B1 (−0.76 kW) becomes a small positive one (+0.21 kW). The 99th-percentile absolute error drops from 14.72 (SPM) to 13.11 kW; the two-stage pipeline keeps the lowest RMSE. On the three clear days of Fig. 4 the median tracks the observation (MAE 0.10–0.13 kW vs. 0.74–1.38 kW for B1) without B1's spurious afternoon dips. On the two most variable days, however, it reproduces ramps late, like persistence, and its +16 min MAE exceeds B1's (3.44 vs. 2.94 and 4.05 vs. 3.12 kW), although its interval covers more observations (75% vs. 62%).

Comparison with B2 and B3. On the cloudy subset (Table II), CSR-MMDiff reduces the CRPS of SPM by 25.4% (uncalibrated: 20.4%) and that of PeEn by 18.1%. PV-MM-Diffusion reports 28.3% over its SPM with 60-step DPM-Solver++ and eight RTX 3090s for training (45.4 s for 10 members), and SkyGPT→U-Net 12.5%. Our relative gain is thus close to the full-size reference at a fraction of the cost,

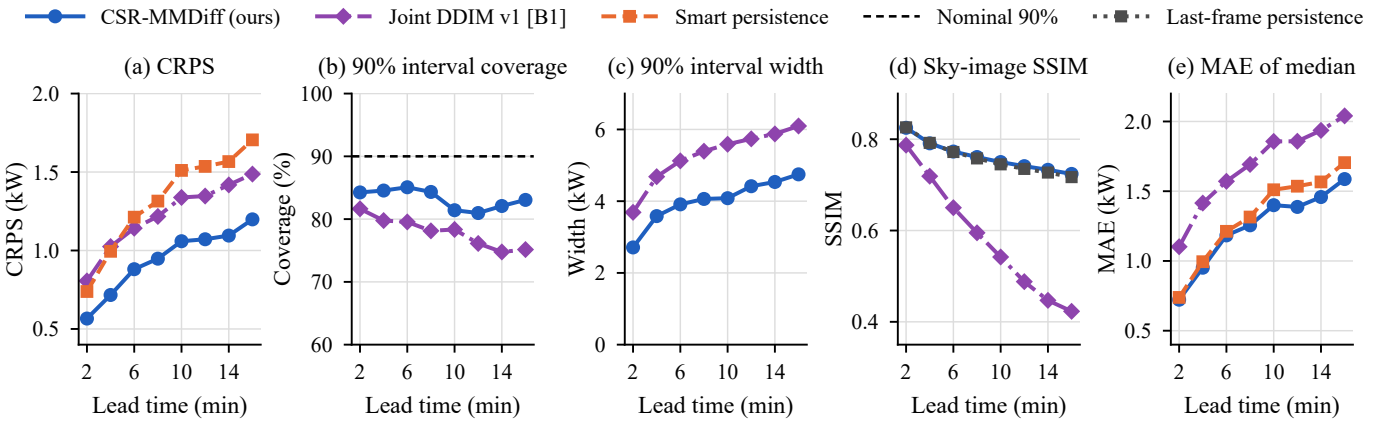


Fig. 2. Performance by lead time on the 1,340 test windows, CSR-MMDiff vs. our earlier joint DDIM model B1 on the same windows. (a) CRPS (MAE for SPM). (b) Coverage and (c) mean width of the 90% interval. (d) SSIM of generated frames (first member) and of last-frame persistence against the observed frames. (e) MAE of the ensemble median.

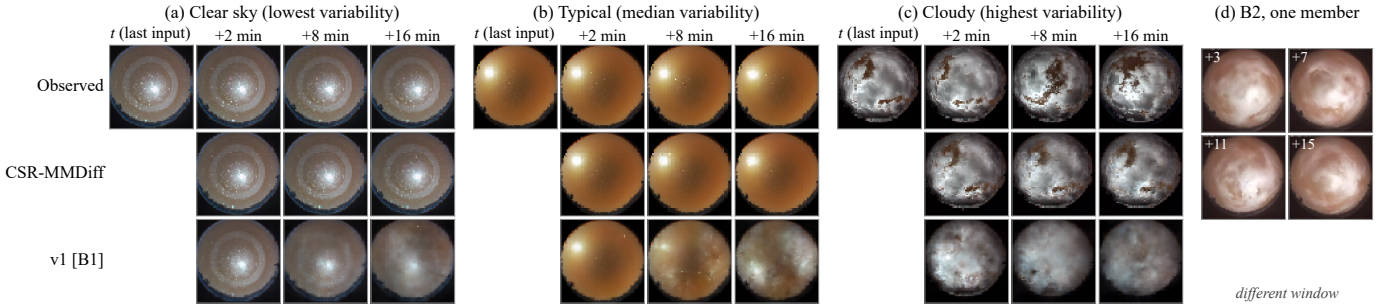


Fig. 3. Observed and generated sky images (first ensemble member) on the lowest-, median- and highest-variability daytime test windows. In (a)–(c), CSR-MMDiff (middle row) and B1 (bottom row) are compared on identical windows and inputs. (d) Generated frames of B2 for another window (one member, lead times relative to its reference time).

but coverage drops to 76.1% on these windows. B2 reached a higher endpoint skill against its own SPM surrogate, with intervals covering only 68–71% of observations; our endpoint coverage is 83.1%.

Generated sky images. The image residual anchors each generated frame to the scene: sun position, lens rings and cloud layout persist (Fig. 3), whereas B1's frames of the same windows degrade into haze. SSIM falls only from 0.825 to 0.724 over 16 min (B1: 0.787 to 0.423), and VGG-CS rises from 0.862 to 0.959 (Table III). Our SSIM, however, stays within 0.01 of last-frame persistence (Fig. 2(d)) and the clouds of the cloudy window barely move, so the PV gain stems from the residual formulation rather than explicit advection. B3's best-of-N VGG-CS (cloudy days, other VGG layer) is indicative only.

Sampler and cost. On 512 validation windows (SPM: 0.832 kW), 5, 10, 20 and 50 DDIM steps give a CRPS of 0.653, 0.640, 0.617 and 0.613 kW at 13, 24, 51 and 134 ms per member. Deterministic sampling ($\eta = 0$) beats $\eta = 0.5$ and 1 (0.612 vs. 0.632 and 0.650 kW), and sky-image guidance hurts ($g = 0.5$: 0.648; $g = 1$: 0.664), so spread is best added by calibration. Test sampling takes 54.1 ms per member on the two RTX 3080s (B1: 155.5 ms on one RTX 3070 at 15 steps).

TABLE III
GENERATED-IMAGE QUALITY (AVERAGED OVER LEAD TIMES).

Method	Sample	VGG-CS	SSIM +2	SSIM +16
Last-frame persistence	–	–	**0.826**	0.717
Joint DDIM v1 B1	member 1	0.862	0.787	0.423
Ours (CSR-MMDiff)	member 1	**0.959**	0.825	**0.724**
DDIM B2	member 1	not evaluated (qualitative only)		
SkyGPT* [11]	best of 10	0.758	–	–
PV-MM-Diffusion* B3	best of 30	0.774	–	–

*From B3's Fig. 5(a): five cloudy test days, leads 1–15 min, different VGG-16 feature layer.

VI. DISCUSSION AND LIMITATIONS

Placing smart persistence inside the diffusion target turns the generative model into a probabilistic *correction* of a physical forecast. This removes the bias of B1 and lets a 5 M-parameter model approach the relative skill of a much larger reference, but the median inherits the lag of persistence on days with frequent ramps, the image branch learns little advection, and coverage is 6.8 points below nominal overall and 13.9 points below on cloudy windows, because one factor per lead cannot adapt to the weather regime. Regime-conditional or conformal recalibration and a longer image context are natural next steps.

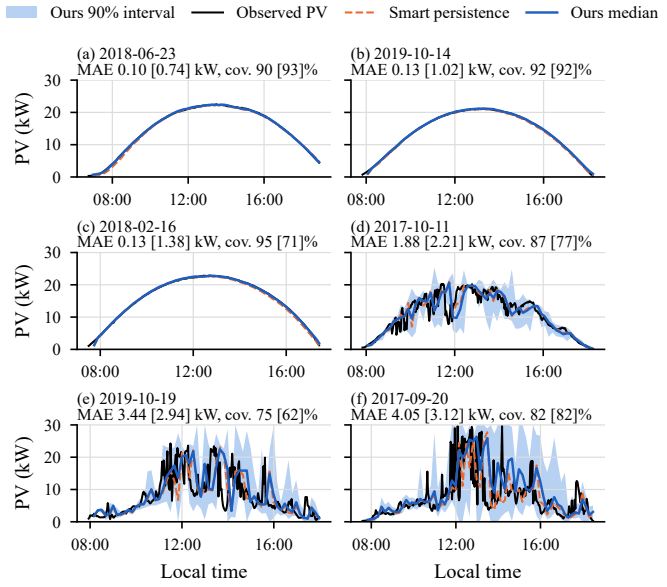


Fig. 4. 16-min-ahead forecasts of CSR-MMDiff on six test days, from least (a) to most (f) variable. Panel titles give the +16 min MAE of the median and the 90% coverage, with B1's values in brackets.

The study covers one site, one horizon, 20 test days and one training run, and B2 and B3 use other test protocols.

VII. Conclusion

Diffusing sky-image and PV residuals with respect to a clear-sky persistence prior lowers the CRPS of our previous joint model by 22.9% to 0.943 kW (28.7% skill over smart persistence), raises 90% coverage from 77.9% to 83.2% with narrower intervals, beats persistence as a point forecaster and keeps generated frames sharp at 54 ms per member. Closing the coverage gap on cloudy days is the main open problem.

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